

Federated O-RAN Slice SLA Prediction: A Cross-Architecture Empirical Benchmark on Colosseum/ColO-RAN

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Abstract—Federated learning (FL) on O-RAN edge gNBs is the natural paradigm for slice-level SLA-violation forecasting because per-user telemetry cannot be centrally pooled. Despite a decade of FL benchmark research, the deployment design space — sequence model architecture, federation algorithm, client-heterogeneity partition, commodity edge hardware — has been studied one axis at a time, leaving the joint behaviour under realistic O-RAN traffic characterised only anecdotally. This paper presents the first cross-architecture empirical FL benchmark on the publicly-available Colosseum/ColO-RAN testbed: 3 sequence backbones (LSTM, Mamba, Spiking-SSM) $\times$ 5 algorithms (FedAvg, FedProx, FedAdam, SCAFFOLD, FedDyn) $\times$ 6 partitions (natural-by-BS + Dirichlet $\alpha \in \{0.05, 0.10, 0.50, 1.00, 5.00\}$) $\times$ 10 seeds = 900 cells, with NVML idle-baseline-subtracted training-energy measurement on RTX 4080 commodity hardware, augmented by a 15-cell controlled random-split ablation on 4 $\times$ Tesla V100-SXM2-32GB to probe the leading mechanism hypothesis. We report four empirical findings: (1) client heterogeneity in O-RAN is structurally helpful, not harmful — the natural base-station partition uniformly outperforms every parametric Dirichlet α across all 3 architectures $\times$ 5 algorithms, inverting the standard FL-benchmark assumption; (2) the algorithm-design space is flat at FedAdam-saturated headroom ($\leq +0.016$ AUC over FedAvg), bounding the room available to recent sharpness-aware methods; (3) selective-state-space backbones interact catastrophically with control-variate algorithms at moderate partition skew (Mamba $\times$ SCAFFOLD $\times$ $\alpha \in \{0.10, 0.50\}$); (4) architecture choice traverses 10 $\times$ in commodity-GPU energy and an order-of-magnitude larger AUC range than algorithm choice, dwarfing the algorithmic-leverage scale. We provide a reference benchmark with paired-bootstrap CI_{95} over 270 paired distributions, NVML-instrumented per-cell energy, and pre-registered statistical protocol; the artefact is released under Apache-2.0 with Croissant dataset metadata and a Zenodo DOI on paper acceptance.

Index Terms—Federated learning, O-RAN, slice management, SLA prediction, state-space models, spiking neural networks, NVML energy measurement, paired-bootstrap inference.

I. INTRODUCTION

OPEN Radio Access Networks (O-RAN) decomposes the cellular base station into commodity hardware components controlled by ML-driven xApps and rApps running on disaggregated RAN Intelligent Controllers (RICs) [1]. Slice-level Service Level Agreement (SLA) violation forecasting is one near-RT RIC xApp pattern of practical interest: each gNB simultaneously serves heterogeneous traffic slices (the ColO-RAN release exercises eMBB constant-bitrate, MTC

Poisson, and URLLC Poisson — see Section III), and proactive forecasting of next-second SLA risk informs near-RT RIC resource-allocation decisions within its 10 ms – 1 s control loop. Because edge gNBs accumulate per-user telemetry that cannot be centrally pooled, federated learning (FL) is the natural training paradigm: each gNB trains a local sequence model on its own traces and shares only weight updates with the FL aggregator. We place the aggregator in the non-RT RIC / SMO orchestration layer (≥ 1 s timescale fits our 100-round federated training cadence per Section ??); a near-RT-RIC-internal xApp aggregator would alternatively be possible if cross-cell xApp messaging supports the federation protocol — we do not engage with that deployment alternative beyond noting it.

Standard FL practice — codified by a decade of toy-image benchmarks — treats client heterogeneity as a *wound* to be healed: lower Dirichlet α (more skew) drives lower accuracy, motivating sharpness-aware methods, control variates, and proximal regularizers. **In O-RAN slice SLA prediction, this premise inverts.** When clients are the *natural* base stations of a Colosseum/ColO-RAN testbed (one client per gNB), the federated average across 7 base-station-natural clients consistently outperforms every uniform-Dirichlet partition of the same training data on the same model + algorithm. Empirically (Section ??, all values are out-of-sample test AUC averaged over 10 seeds), the natural-by-BS partition achieves AUC 0.913–0.918 on LSTM (5-algo range; $\Delta \approx 0.005$), 0.8998–0.9186 on Mamba (with SCAFFOLD as the lower outlier and the other four algorithms in 0.911–0.919), and 0.840–0.856 on Spiking-SSM, **strictly above the AUC obtained at any parametric Dirichlet α** for every (arch, algo) pair, and monotonically increasing as $\alpha \rightarrow 0$ within the Dirichlet sweep. The $\alpha = 5.0 \rightarrow \alpha = 0.05$ AUC gap is ≈ 0.10 – 0.12 on the dense LSTM/Mamba backbones; on Spiking-SSM the gap is smaller (≈ 0.022) but still positive, and across all five algorithms heterogeneity is associated with higher AUC, not lower. The mechanism is open — natural-by-BS is *not* the low- α limit of the Dirichlet sweep (the two partition families redistribute rows along orthogonal axes, see Section ??) — and we present a controlled `random_split` ablation in Section ?? (15 cells $\times$ 3 architectures on 4 $\times$ Tesla V100-SXM2-32GB) showing that any partition that breaks bs grouping (whether by per-slice Dirichlet redistribution at $\alpha = 5.00$ or by uniform random shuffling) collapses AUC to a tight band ~ 0.18 below natural-by-BS, and is statistically equivalent to Dirichlet $\alpha = 5.00$ within seed σ . The mechanism direction (preservation of bs

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grouping) is therefore confirmed; the precise dataset axis the model exploits via bs grouping (channel-state features, see Section ??) remains under further investigation.

Despite the explosion of FL benchmarks since 2018, the deployment design space for FL-based slice SLA prediction remains under-characterized. The space spans four axes:

- 1) **Sequence model architecture** — recurrent (LSTM), structured state-space (Mamba), or bio-inspired sparse spiking variants (Spiking-SSM).
- 2) **Federation algorithm** — FedAvg, FedProx, FedAdam, SCAFFOLD, FedDyn (and the more recent sharpness-aware family discussed in Section II).
- 3) **Client heterogeneity** — including the natural base-station partition (one gNB $\Rightarrow$ one client) and parametric Dirichlet stress $\alpha \in \{0.05, 0.10, 0.50, 1.00, 5.00\}$ over slice IDs.
- 4) **Commodity hardware** — consumer-tier edge GPUs (RTX-class) at the gNB rather than data-center accelerators.

Existing benchmarks address each axis in isolation. NeurIPS-class FL benchmarks evaluate algorithm $\times$ heterogeneity on toy image data [2], [3], [4], [5]. Spiking-NN benchmarks compare architectures in centralized regimes on neuromorphic-friendly datasets [6]. FL $\times$ wireless papers focus on a single architecture [7], [8], [9]. The implicit assumption underlying single-axis recommendations: **best architecture $\times$ best algorithm $\times$ representative $\alpha \times$ commodity hardware = best deployment**.

This paper tests the assumption empirically. We construct the first comprehensive 4-axis FL benchmark on the publicly-available Colosseum/CoLO-RAN testbed dataset [1], [10], sweeping $\{\text{LSTM, Mamba, Spiking-SSM}\} \times \{\text{FedAvg, FedProx, FedAdam, SCAFFOLD, FedDyn}\} \times \{\text{natural-by-BS, Dirichlet } \alpha \in \{0.05, 0.10, 0.50, 1.00, 5.00\}\} \times 10 \text{ seeds} = 900 \text{ cells (90 groups, all 10 seeds populated), with NVML-instrumented per-round GPU energy measurement on RTX 4080 (sm_89) commodity hardware. Statistical inference uses paired-bootstrap } CI_{95} \text{ on per-seed AUC deltas; Wilcoxon } p\text{-values are reported uncorrected per finding (with family-wise Bonferroni discussed in Section ?? as a supplementary check).}$

Contributions:

- 1) **Inverted heterogeneity in O-RAN slice federation:** against the standard FL-benchmark assumption, the natural base-station partition uniformly outperforms every parametric Dirichlet α in our sweep across all 3 architectures and 5 algorithms. We do not claim a closed-form mechanism — Section ?? reports a controlled `random_split` ablation that tests the leading “bs-conditioned signal preservation” hypothesis — but the empirical pattern is the first published characterization on Colosseum/CoLO-RAN and is deployment-relevant regardless of which mechanism candidate ultimately survives.
- 2) **Algorithm-design space is flat in this regime; FedAdam achieves the empirical maximum in our 5-algorithm baseline:** across Dirichlet $\alpha \in \{0.05..5.0\}$,

the spread between the best and worst of $\{\text{FedAvg, FedProx, FedAdam, SCAFFOLD, FedDyn}\}$ on a fixed (arch, partition) cell is 0.022–0.031 AUC on LSTM/Spiking and 0.078 on Mamba (driven by a single SCAFFOLD outlier). FedAdam consistently leads FedAvg via paired-bootstrap CI_{95} (LSTM Dirichlet-averaged $\Delta = +0.0080$, range $+0.0048.. +0.0128$, all 5 cells significant; Mamba Dirichlet-averaged $\Delta = +0.0062$, range $-0.0021.. +0.0097$, 4 of 5 cells significant; Spiking Dirichlet-averaged $\Delta = +0.0164$, range $+0.0108.. +0.0267$, 4 of 5 cells significant). The mechanism analysis in Sections II-F + ?? + ?? argues that LookAhead-style server-side aggregation (FedSWA / FedSCAM) is *unlikely to systematically exceed* this empirical maximum because of the variance-estimation gap between Adam-style and LookAhead-style server steps (predictive claim, not theorem); we accordingly dismiss FedSWA [11] and FedSCAM [12] via mechanism analysis rather than benchmark them. We do not benchmark FedBN [13] either; Section ?? L2 discusses why the algorithm-flatness ranking is unlikely to be overturned by FedBN given its reported gain envelope on related cellular feature-skew tasks.

- 3) **Architecture $\times$ algorithm catastrophic interaction (Mamba $\times$ SCAFFOLD $\times \alpha \in \{0.10, 0.50\}$):** SCAFFOLD on Mamba at $\alpha = 0.10$ yields AUC 0.7609 ± 0.0830 — std amplified $\sim 3\times$ over (Mamba, FedAvg) at the same α (0.0251) and $\sim 10\times$ over (Mamba, FedAvg) at $\alpha = 5.0$ (0.0073). The deployment warning is concrete: control-variate methods interact pathologically with selective-scan SSMs under high heterogeneity, even as both components individually behave well on neighboring cells.
- 4) **Architecture leverage dominates algorithm leverage (paired-bootstrap CI_{95}) on RTX 4080:** on the energy-AUC Pareto front (Section ??), architecture choice spans $10\times$ in NVML-measured model-attributable energy on RTX 4080 (LSTM $\sim 3.5\text{ kJ} \rightarrow$ Mamba $\sim 10.5\text{ kJ} \rightarrow$ Spiking $\sim 41\text{ kJ}$ per cell). Holding (algorithm, IID partition) fixed, the paired Spiking – LSTM AUC delta is concentrated at -0.061 to -0.074 across all 5 algorithms (all 5 cells significant at $p < 0.005$), substantively dwarfing the FedAdam – FedAvg algorithmic gap of $+0.006$ to $+0.016$ on the same architectures (contribution 2). The $10\times$ span is hardware-specific: the same architectures on a sparsity-aware accelerator (Loihi-2, IBM NorthPole) would yield a substantially smaller LSTM-vs-Spiking energy ratio (Section ?? + Section ?? L1, L6). Operator decisions on commodity edge GPU should center architecture-energy trade-off, not algorithm breadth.
- 5) **Open reproducible reference benchmark:** spec-driven YAML pipeline with pre-flight algorithm validation and `--skip-completed` resumability; paired-bootstrap statistical pipeline; NVML idle-baseline-subtracted training energy; Croissant metadata; Zenodo DOI on acceptance; Apache-2.0 license (preregistered). The full 4-axis table maps deployment configurations $\rightarrow$ (AUC, F1, energy,

paired CI_{95}) over 90 (arch $\times$ algorithm $\times$ partition) groups, enabling lookup-style operator decisions on Colosseum/ColO-RAN.

The remainder of the paper is organized as follows. Section II reviews related work across the four axes. Section III describes the Colosseum/ColO-RAN dataset and our preprocessing pipeline. Section ?? details the architecture, algorithm, and partition methodology. Section ?? enumerates the reproducibility infrastructure and our Croissant-aligned data release. Section ?? reports results across the 90 (arch $\times$ algorithm $\times$ partition) groups with paired-bootstrap CI_{95} . Section ?? discusses the mechanism finding and articulates the applicability boundary on commodity edge hardware. Section ?? enumerates limitations. Section ?? concludes.

II. RELATED WORK

A. Federated learning benchmarks

The canonical FL benchmark is **LEAF** [2], which standardized federated MNIST/Shakespeare evaluation pipelines for early FedAvg variants. **Flower** [14] later provided a framework rather than a benchmark, supporting heterogeneous device experiments. **FedScale** [3] scaled to thousands of cross-device clients; **FLAIR** [4] introduced a long-tail multi-label image dataset. **pfl-research** [5] integrates differential-privacy mechanisms into a simulation framework. Most recently, **few-bench** [15] introduced rigorous bootstrap- CI_{95} evaluation for time-series forecasting, though without the federated dimension. None of these benchmarks targets O-RAN slice SLA prediction specifically; their toy-data settings cannot capture the covariate-skew structure of real RAN telemetry.

B. FL on cellular and wireless networks

A first generation of FL $\times$ cellular work targeted SLA-constrained slicing under the Beyond-5G banner. **Statistical FL for B5G SLA-Constrained RAN Slicing** [16] introduced a long-term CDF SLA constraint. **CDF-Aware FL** [17] formalized the constraint as a per-client penalty. **Fed-LSTM** [7] deployed an LSTM forecaster for slice-level traffic. **TRACTOR** [18] integrated reinforcement-learning xApps for resource-block assignment. **REAL** [19] integrated the OSC near-RT RIC with srsRAN for RL-based slice-resource-allocation xApps under urban-mobility channel modelling. Recently **Hayek et al. 2025** [20] measured FL convergence on a 5G/WiFi/Ethernet COTS testbed with Raspberry-Pi clients, exposing uplink straggler effects. **arXiv:2508.08479** [8] benchmarked FedAvg/FedProx/FedBN on five throughput-prediction datasets with LSTM, CNN, CNN+LSTM, and Transformer architectures, finding FedBN superior under feature skew. **Mangi et al. 2026** [9] (“WHEN CLIENTS DRIFT”) proposed regime-aware aggregation under leave-one-regime-out evaluation on synthetic 6G RAN telemetry.

Each of these papers covers one or two of our four axes. None benchmarks LSTM, Mamba, and Spiking-SSM jointly under parametric Dirichlet heterogeneity, and none reports per-round NVML energy. Our work is differentiated by the combination of architecture breadth, algorithm breadth, parametric

heterogeneity, and energy instrumentation on the **publicly-available** Colosseum/ColO-RAN dataset, in contrast to closed simulators or undisclosed datasets used by Mangi et al.

C. Spiking-SSM and FL $\times$ spiking neural networks

Hybrid spiking-state-space architectures emerged from late 2024. **SpikMamba** [21] combined LIF spiking neurons with Mamba selective scan for event-camera action recognition. **Mamba-Spike** [22] introduced a spiking front-end for general temporal data. **Spiking Point Mamba** [23] extended to 3D point clouds. **SpikingSSMs** [24] formalized sparse-parallel spiking state-space layers. **SpikingMamba** [25] distilled large language models into spiking variants for energy efficiency. **SpikingBrain** [26] scaled spiking architectures to 76B parameters on data-center GPU clusters with explicit FLOP-utilization measurement.

Federated-learning variants of spiking neural networks emerged earlier than spiking-SSM: **arXiv:2106.06579** [27] founded FL $\times$ SNN. **FedLEC** [28] extended to label-skew non-IID. **SNN in vertical FL** [29] examined VFL energy trade-offs. **Robustness of SNN in FL with compression** [30] addressed Byzantine attacks. **Privacy in FL with SNN** [31] characterized gradient leakage. Most recently, **arXiv:2602.12009** [32] examined firing-rate sensitivity to differential privacy in federated SNN — preempting the DP-SNN-FL angle. We accordingly cite this work in Section ?? as a deferral and do not duplicate the DP study; our contribution is orthogonal in its multi-architecture $\times$ parametric-Dirichlet $\times$ NVML scope.

To our knowledge, no published work combines spiking-SSM with FL on cellular/RAN telemetry. Our `spiking_expand2` variant adapts the wide-inner-state design pattern from Mamba [33] to spiking-SSM and demonstrates competitive AUC/energy trade-off on commodity edge GPU under FL.

D. GPU energy measurement methodology

The argument that FLOP counts mis-estimate actual hardware energy predates our work. **Shen et al. 2023** [34] (“Is Conventional SNN Really Efficient?”) critiqued synaptic-operation accounting via a Bit Budget framework, finding that quantized ANNs dominate SNNs at equivalent bit budgets. **α -FLOPs** [35] proposed a parallelism-aware alternative. **NeuroBench** [36] standardized neuromorphic benchmarking on real hardware. The **ML.ENERGY Benchmark** [37] measured inference energy on generative AI; **Where Do the Joules Go?** [38] diagnosed the breakdown across model families. **STEP** [6] specifically benchmarked spiking transformers including memory-access cost. **Beyond Backpropagation** [39] applied NVML and CodeCarbon to alternative training objectives.

We follow the **Energy API** approach (`nvmlDeviceGetTotalEnergyConsumption`, available on Volta+ with NVML ≥ 11.0) preferred over polling-power Riemann-sum integration per the ML.ENERGY 2025 paper [37] and accompanying leaderboard. Our novel angle is the **federated-training regime**: prior NVML benchmarks

measure inference [36], [37] or centralized training [26], [39]; we measure per-round energy across an FL sweep, including idle-baseline subtraction.

E. Heterogeneity in federated learning

NIID-Bench [40] established Dirichlet partitioning as the standard non-IID benchmark. **FedBN** [13] keeps client-local BatchNorm statistics for feature-skew settings; **arXiv:2508.08479** [8] demonstrated FedBN’s superiority on cellular feature-skew. **pFedFDA** [41] specifically targets covariate shift. **Borazjani et al. 2025** [42] argued for embedding-based heterogeneity definitions beyond label-skew.

We use Dirichlet partitioning over slice IDs as the canonical covariate-skew model for O-RAN: each gNB serves a different mix of slice traffic, with mixing entropy controlled by α_{dir} . We adopt $\alpha_{\text{dir}} \in \{0.05, 0.10, 0.50, 1.00, 5.00\}$ to span extreme stress (each gNB dominated by a single slice) through near-IID. We additionally evaluate the natural base-station partition (one gNB $\Rightarrow$ one client, 7 CoLo-RAN gNBs in the public release) as a non-parametric heterogeneity reference. We discuss the applicability of embedding-based alternatives in Section ??.

F. Sharpness-aware federated learning (2024–2026)

A separate strand of recent FL work targets *heterogeneity-induced sharp minima* via per-step learning-rate cycling and server-side weight extrapolation. **FedSAM** [43] adapted Foret et al.’s SAM perturbation to the federated client step. **FedSWA** [11] introduced an intra-round cyclical local LR schedule (eq. 3) combined with a LookAhead-style server EMA $\theta_t = \theta_{t-1} + \alpha(v_t - \theta_{t-1})$ with $\alpha = 1.5$ (eq. 17). **FedSCAM** [12] composed SAM with clustering for further sharpness regularization. The reported gains (FedSWA: +5.6% over FedAvg on CIFAR-100 with 10% client participation across 1000 rounds) assume large- N low-participation FL with coherent client drift toward sharp minima.

We do not empirically evaluate this family on Colosseum/CoLo-RAN (see Section ??). Two structural mismatches make the absence of comparison defensible on mechanism alone. First, our regime — 7 base-station-natural clients with 71% per-round participation, 100 rounds, RAN slice SLA prediction — is *aggregation-noise-dominated* rather than coherent-drift-dominated. Second, FedAdam’s adaptive variance damping Adam(m, v) over $(v_t - \theta_{t-1})$ systematically dominates FedSWA’s variance-blind LookAhead overshoot *in expectation* in this regime: FedAdam knows when to step small (high v) and when to step normally (low v), while FedSWA’s LookAhead extrapolation rate α_{LA} (the FedSWA paper’s α -symbol; we write it α_{LA} throughout to disambiguate from the Dirichlet α_{dir} of Section ?? and the FedDyn α_{dyn} of Section ??; FedSWA’s preregistered value is $\alpha_{\text{LA}} = 1.5$) amplifies *both* signal and noise uniformly. Empirically (Section ??), FedAdam already saturates the headroom available to server-side aggregation modifications at +0.006–+0.016 Dirichlet-averaged AUC over FedAvg across the 3 architectures (paired-bootstrap CI_{95} in Section I contribution 2); LookAhead overshoot is unlikely to *systematically*

TABLE I
COLO-RAN SLICE AND SCHEDULER INDEX MAPPING.

Index	Slice ID	Traffic class	Workload
0	eMBB	broadband	4 Mbps CBR/UE
1	MTC	machine-type	Poisson 30 pkt/s, 125 B/UE
2	URLLC	ultra-reliable	Poisson 10 pkt/s, 125 B/UE
Index	sched ID	Scheduler	
0	RR	Round-Robin	
1	WF	Waterfilling	
2	PF	Proportionally Fair	

exceed this ceiling without the variance-estimation machinery FedAdam has and FedSWA lacks. We additionally note that FedSWA targets *heterogeneity-as-sharpness-wound*, which our inverted- α finding (Section I, Section ??) indicates does not characterize this dataset.

III. DATASET

A. Colosseum / CoLo-RAN testbed

CoLo-RAN [1] is the publicly-released RAN telemetry corpus collected from the Colosseum digital-twin testbed [10], comprising the per-second downlink/uplink physical-layer measurements of 7 base stations (Colosseum Rome scenario, BS nodes {1, 8, 15, 22, 29, 36, 43}; we re-index these to `bs_id` $\in \{1, \dots, 7\}$ for ease of use) operating concurrently under 28 distinct traffic configurations (`tr` $\in \{0, \dots, 27\}$, each a different RBG-per-slice allocation pattern; the exact (slice 0 / slice 1 / slice 2) RBG triples per `tr` are documented in the CoLo-RAN release). Each base station serves 3 logical slices, and each slice can be served by one of 3 scheduling policies; both index mappings are listed in Table I. The unified telemetry table contains ≈ 18 M rows of (timestamp, `bs_id`, `slice_id`, `sched`, `tr`) keyed observations with 17 continuous features (uplink/downlink throughput, MCS/CQI, BLER, RB utilisation, buffer occupancy; full list in Section ??). We use the public release without modification; no synthetic augmentation, no re-simulation.

B. Task definition: per-slice SLA-violation forecasting

The forecasting target is the binary indicator $y_{t+1} = 1[\text{ul_bler}_{t+1} > 0.10]$, predicting whether the next-second uplink BLER on a given (`bs_id`, `slice_id`) tuple will breach the 10% SLA threshold. This threshold is the canonical SLA-violation gate used by the CoLo-RAN authors [1]; the binary framing keeps the metric (AUC, F1) decision-relevant for the near-RT RIC’s resource-allocation xApps. The empirical positive rate on the train partition is 30.9% (per-slice 34.9% / 26.7% / 31.0%, measured on `tr` $\in \{0..21\}$; see Section ?? for how this rate enters the per-client loss reweighting). Inputs to the model are the most recent five seconds of (`slice_id`, `sched`, `tr`, 17 continuous features) on the same (`bs_id`, `slice_id`) tuple — `seq_len=5` is preregistered from v3/v4.

C. Out-of-distribution split by traffic config

To prevent label leakage between train / validation / test, we partition the 28 traffic configurations into three disjoint groups:

train $\text{tr} \in \{0..21\}$ (22 configs), validation $\text{tr} \in \{22..24\}$ (3 configs), and test $\text{tr} \in \{25..27\}$ (3 configs). All three sets are evaluated on the same 7 base stations and 3 slices; only the traffic-config index changes. This split is identical to v4 so accuracy numbers are cross-paper comparable. Crucially, **partition-into-clients (Section ??) operates within the train tr range only**: validation and test sets are pooled across all clients to guarantee identical evaluation surfaces across configurations.

D. Target-leakage audit (preserved from v4)

A v4-era audit established that the previously documented `allocation_efficiency = 0.5 * throughput_eff + 0.3 * qos + 0.2 * prb_util` regression target is a deterministic linear combination of three concurrent inputs and is therefore unsuitable. We retain that exclusion and the v4-validated leak-free continuous feature set in v7.

E. Preprocessing pipeline

The 18M-row unified parquet (`coloran_raw_unified.parquet`) is materialised once and streamed lazily through four pipeline stages: (i) `per(bs_id, slice_id)` sequence construction with `build_run_sequences` (rolling `seq_len=5` windows, positive-rate computed from train rows only via `pos_weight_split=train` (preregistered) — this avoids test-set leakage into the loss function); (ii) leak-free `ContinuousScaler` fit on the train partition only and applied to validation and test; (iii) categorical encoding of `slice_id × sched × tr` via `nn.Embedding` lookups; (iv) Dirichlet-or-natural-by-BS client partition (Section ??). All four stages are implemented as pure functions in `src/fl_oran/data_v2/` and tested in `tests/test_v5_*` for determinism and seed-reproducibility (single-source-of-truth contract). The same sequence index is shared across all algorithms within an (arch, partition, seed) triple via a `SharedSplits` cache; this is the perf inheritance from prior centralized work.

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